

“CONVEX OR CONICAL HEAPS, INCREASING UPWARD FROM A HORIZONTAL BASE.”

Luke Howard, British meteorologist, describing cumulus clouds in *Essay on the Modifications*, 1803

British pharmacist and meteorologist Luke Howard derived the term cumulus from the Latin word for “heap.” As seen in this image, cumulus clouds appear cottonlike and tend to have flat bases.

ON MARCH 28, 1803, THE CHARLOTTE DUNDAS, designed by British engineer **William Symington** (1764–1831), became the **first practical steamboat** when it towed two laden barges through the Forth and Clyde Canal in Scotland.

The following year, another British engineer, **Thomas Telford** (1757–1834), began work on the **Caledonian Canal** in Scotland. When completed in 1822, this canal was 60 miles (100 km) long and 101 ft (30.5 m) wide, with 28 locks—stretches of water enclosed by gates—

that were the largest in the world at the time.

In October 1803, British chemist **John Dalton** presented his **atomic theory** to an audience in Manchester, UK. He suggested that: all matter is composed of atoms; atoms cannot be made or destroyed; all atoms of the same element are identical; and different elements have different types of atoms. Dalton also stated that chemical reactions occur when atoms are rearranged, and that compounds are formed from atoms of the constituent elements.

Also in 1803, British scientist **William Hyde Wollaston** added to the number of known elements by discovering **palladium** and **rhodium**.

British pharmacist and meteorologist **Luke Howard** (1772–1864) published a **description of clouds**. He used Latin names to classify

Dalton's table of elements
John Dalton was the first to use symbols for elements, and to calculate their atomic weights. This table shows his symbols and atomic weights for 20 elements.

ELEMENTS			
○	Hydrogen.	☼	Strontian
⊖	Nitrogen	⊗	Barytes
●	Carbon	⊙	Iron
○	Oxygen	⊕	Zinc
⊕	Phosphorus	⊖	Copper
⊗	Sulphur	⊙	Lead
⊕	Magnesia	⊖	Silver
⊗	Lime	⊙	Gold
⊕	Soda	⊖	Platina
⊗	Potash	⊙	Mercury

them into three simple categories—cirrus, cumulus, and stratus—that are still used.

On February 21, 1804, 25 years before Stephenson's *Rocket* (see 1829), a **steam locomotive** designed by British engineer **Richard Trevithick** (1771–1833) hauled 70 passengers, 11.2 tons (10 tonnes) of iron, and five wagons from the ironworks at

12 PERCENT THE AMOUNT OF MORPHINE THAT OPIUM CAN CONTAIN

Penydarren to the Merthyr-Cardiff Canal in Wales, a distance of 9 miles (14 km). It reached a speed of nearly 5 mph (8 kmph).

Swiss chemist **Nicolas-Théodore de Saussure** (1767–1845) outlined the process of **photosynthesis** (see 1787–88) and proved that both water and carbon dioxide are absorbed by plants as they grow. Saussure later analyzed the ashes of plants and showed that their

Surgical breakthrough
Hanaoka Seishū carried out the first successful surgery using general anesthesia on a 60-year-old woman who suffered from breast cancer.

mineral composition differed from that of soil, indicating that plants absorbed nutrients selectively.

German pharmacist **Friedrich Sertürner** (1783–1841) became the first person to isolate the active ingredient of a medicinal plant. In experiments starting in 1803 and published in 1805, he isolated **morphine** from opium. This substance would later become invaluable in surgery.



Japanese surgeon **Hanaoka Seishū** (1760–1835) performed the first successful surgery using **general anesthesia** in October. The anesthetic was an orally administered herbal concoction.

JOHN DALTON (1766–1844)

A Quaker schoolteacher, John Dalton was secretary of the Manchester Literary and Philosophical Society, UK, from 1800. Best remembered for his theory of atoms, Dalton also contributed to meteorology and studied color blindness, from which he suffered. He made meteorological observations and published scientific papers well into his seventies.

